

Replicating Manifold Steering: Geometry of LLM Representations and Behavior

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Code, checkpoints, and figures: <https://github.com/kirillacharya/manifold-steering-replication>

Why geometry?

Large language models build up internal representations layer by layer: at layer L , each token maps to a hidden state $h^{(L)} \in \mathbb{R}^d$, and structured concept domains may organize into geometric patterns. Hidden states for *days of the week* or *months of the year* may trace a smooth, low-dimensional manifold $\mathcal{M}_h \subset \mathbb{R}^d$ — closed loops for cyclic domains, open curves for sequential ones. Wurgaft et al. [1] ask whether this geometry *predicts behavior*: output probability vectors, embedded via the Hellinger map $p \mapsto \sqrt{p}$ (which makes the Hellinger distance $d_H(p, q) = \frac{1}{\sqrt{2}} \|\sqrt{p} - \sqrt{q}\|$ Euclidean [2]), may trace a matching *behavior manifold* $\mathcal{M}_y$. If $\mathcal{M}_h \approx \mathcal{M}_y$ isometrically, steering *along* $\mathcal{M}_h$ should produce natural behavioral transitions, while linear steering cuts off-manifold and “teleports” probability mass. The two strategies interpolate between a source concept $\hat{h}_{c_s}$ and a target $\hat{h}_{c_e}$:

$$\pi_{\text{lin}}(\alpha) = (1 - \alpha) \hat{h}_{c_s} + \alpha \hat{h}_{c_e}, \quad \pi_{\text{mfd}}(\alpha) = s((1 - \alpha) u_{c_s} + \alpha u_{c_e}), \quad u_{c_i} = s^{-1}(\hat{h}_{c_i}),$$

where $s : \mathbb{R}^k \rightarrow \mathbb{R}^d$ maps intrinsic coordinates back to activation space. We replicate the paper’s central figure on a small open model, then extend it to a 2D concept structure (a day×hour torus).

The experiment, step by step

Setup. All experiments use instruction-tuned **Gemma-2-2B** ($d = 2304$), patching layer 24 of 26. Two cyclic concept domains: *weekdays* (7 classes) and *months* (12 classes), probed with arithmetic prompts — “Q: What day is two days after Monday? A:” (answer: *Wednesday*), 6 prompts per concept (`scripts/run_weekdays.py`, `run_months.py`). For each prompt we record the last-token hidden state $h^{(24)}$ and the output distribution over concept tokens, then

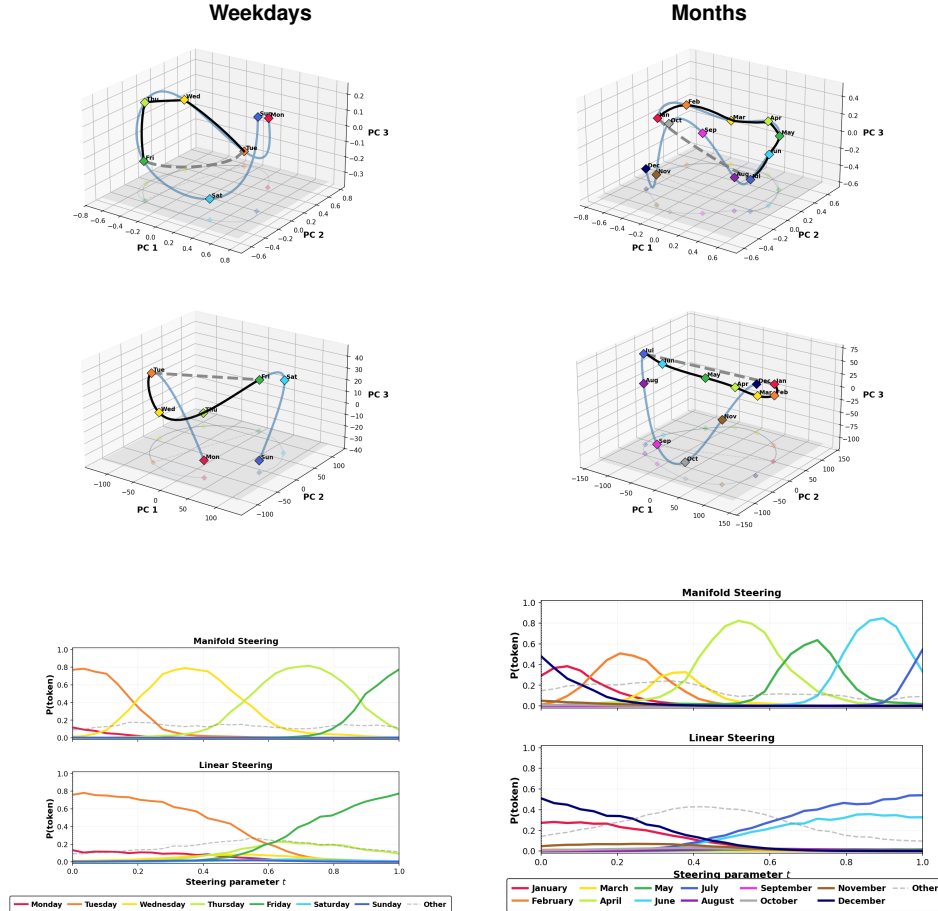


Figure 1: Gemma-2-2B layer 24, steering Tue→Fri (weekdays) and Jan→Jul (months). Rows 1–2: fitted behavior manifolds $\mathcal{M}_y$ (Hellinger–PCA) and activation manifolds $\mathcal{M}_h$ (activation–PCA) with the linear path (dashed) and manifold path (black). Row 3: concept-token probabilities along each path.

average per concept: $\hat{h}_c = \frac{1}{N_c} \sum_{i \in S_c} h_i^{(L)}$ and $\hat{p}_c = \frac{1}{N_c} \sum_{i \in S_c} p_i$.

Manifolds. Activation centroids $\{\hat{h}_{c_k}\}$ are reduced by PCA to r dimensions ($r=6$ weekdays, $r=11$ months) and a cubic spline γ with C^2 continuity is fitted through them in concept order, giving $\mathcal{M}_h$; the same procedure on Hellinger-embedded behavior centroids $\{\sqrt{\hat{p}_{c_k}}\}$ gives $\mathcal{M}_y$.

Steering. For $\alpha \in [0, 1]$ between spline indices i_s, i_t : $\pi_{\text{mfd}}(\alpha) = \text{PCA}^{-1}(\gamma(i_s + \alpha \Delta i))$. At each of 30 steps, a forward hook replaces the layer-24 hidden state at the last token with $\pi(\alpha_t)$ and the output probability of every concept token is recorded (`scripts/run_steering_probs.py`). Everything is deterministic; all figures regenerate from committed checkpoints without a GPU (see the repo README).

Replication of the main figure

Figure 1 replicates the paper’s main result. In both domains the concept centroids trace closed, ordered loops in activation space and in behavior space — the cyclic structure of weeks and months is visibly encoded. The manifold path stays on the fitted curve and, in the probability trajectories (row 3), produces *smooth, ordered* transitions: steering from Tuesday to Friday passes probability mass through Wednesday and Thursday in sequence. Linear steering instead leaves the manifold and *teleports*: intermediate concepts never rise in order, and off-concept mass appears mid-path. One practical lesson: the patch must be applied deep enough. At layer 12, keys and values computed from the original prompt override the patched query over the 14 remaining blocks and both methods flatline; at layer 24 (92% depth) the replication is clean.

Extension: torus steering (day $\times$ hour)

Do LLMs encode *two-dimensional* concept structure? A weekday lives on one circle; a (day, hour) pair lives on a *product* of two circles — a torus. We collect layer-24 activations for all 168 prompts “It is HH:00 on {Day}” (7 days $\times$ 24 hours) and assign each prompt two angles, $\theta_{\text{day}} = 2\pi k/7$ and $\theta_{\text{hour}} = 2\pi h/24$. Each circle’s 2D subspace $V_{\text{day}}, V_{\text{hour}} \in \mathbb{R}^{2 \times d}$ is recovered by regressing centered activations onto the standardized labels $[\sin \theta, \cos \theta]$ — each label component yields one direction $d \propto \sum_i (h_i - \bar{h}) \ell_i$ — followed by QR-orthogonalization of the two directions (`scripts/run_hierarchical_steering.py`). The two recovered planes are nearly orthogonal to each other (max $|\cos| \approx 0.18$), so day and hour occupy largely independent planes of the residual stream. Projecting each activation onto both planes gives two recovered angles, rendered on a torus — day on the major circle, hour on the minor (Fig. 2, left). The 7 day-arcs are cleanly ordered; the hour circle does not fully close, so hour is only partially circular here.

Steering in *product coordinates* respects this factorization: starting from the source activation h_{base} (Tue, 09:00), each step moves the day and hour coordinates *independently within their own planes*, $h(\alpha) = h_{\text{base}} + \Delta_{\text{day}}(\alpha) V_{\text{day}} + \Delta_{\text{hour}}(\alpha) V_{\text{hour}}$, where $\Delta_c(\alpha)$ interpolates the source $\rightarrow$ target offset within each plane — the 2D analogue of walking along the manifold. Each patched activation is decoded into a day $\times$ hour probability grid: day from token probabilities, hour by nearest centroid in the V_{hour} plane (two-digit hours are multi-token for Gemma, so subspace readout is more faithful). Steering (Tue, 09:00) $\rightarrow$ (Fri, 15:00) over six steps (Fig. 2, right), the bright cell walks diagonally — both coordinates move smoothly — while linear steering moves the day correctly but teleports the hour, jumping to 15:00 only at the final step: the same off-manifold failure mode, now visible per coordinate.

Takeaway. Representation geometry is not decoration: moving along it produces natural behavior, and ignoring it breaks transitions exactly as the manifold picture predicts. Code, checkpoints, SLURM scripts, and a no-GPU figure regeneration path: <https://github.com/kirillacharya/manifold-steering-replication>.

References [1] D. Wurgaft, C. Rager, M. Kowal, et al. Manifold steering reveals the shared geometry of neural network representation and behavior. *arXiv:2605.05115*, 2026. [2] S. Amari and H. Nagaoka. *Methods of Information Geometry*. AMS, 2000.

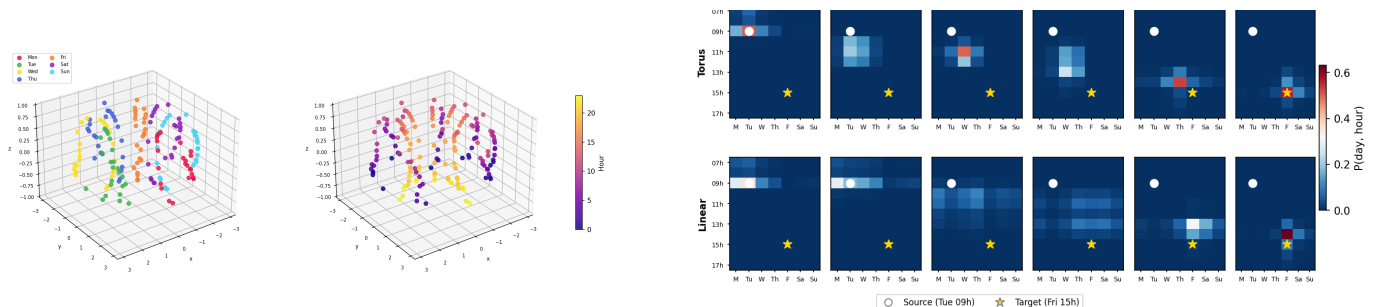


Figure 2: Left: layer-24 activations on the day $\times$ hour torus, colored by day (7 ordered arcs) and by hour. **Right:** day $\times$ hour probability grids across 6 steering steps, (Tue, 09:00) $\rightarrow$ (Fri, 15:00); top row torus (product-coordinate) steering, bottom row linear steering.